

to India's first operational quantum computer and a Quantum Research Park, making the city a key hub for quantum innovation.

The mix of open source platforms, government programs and startup innovation is building a strong Indian ecosystem for quantum technologies. National missions and local collectives such as QCI ensure that training, industry use and homegrown research move forward together. This transformation is being driven by open collaboration, which is turning today's learners into the next generation of quantum experts.

Core algorithms and applications

Any machine learning system depends on algorithms as its core foundation. The process of teaching computers to learn from data requires sequential instructions, which form the basis of these algorithms. Quantum machine learning (QML) uses quantum mechanics to develop new methods for solving problems.

The Quantum Support Vector Machine is a good example in this field of research. Let's take one example to understand it intuitively. A crowded hall requires a method to distinguish between two distinct groups of people. A classical computer performs individual assessments of all possibilities while creating a boundary line. The quantum system performs multiple possibility tests at once to discover the optimal boundary in a very short time. This system proves highly effective for fast classification

operations, which include fraud detection and medical diagnosis.

The Variational Quantum Eigensolver or VQE functions as a core algorithm that determines the best solution from multiple optimisation problem choices. The VQE algorithm provides chemists with various ways to study molecular behaviour.

The Quantum Approximate Optimization Algorithm (QAOA) solves complex problems that include routing and scheduling operations. The system produces flawless delivery routes for thousands of trucks.

Today quantum machine learning applications are transitioning from laboratory environments to real-world deployment. Roche and Cambridge Quantum are working together to apply QML for initial drug development through their research activities. Goldman Sachs and JPMorgan Chase apply QML to perform portfolio risk assessments. Volkswagen leverages QML to forecast traffic movement patterns and Airbus employs quantum techniques to develop lightweight aircraft materials. The Tata Institute of Fundamental Research in India is advancing materials science research through quantum algorithm development.

What's holding QML back?

Despite the excitement, QML is facing numerous obstacles. The most obvious is hardware. The current quantum computers are noisy and their qubit capacity remains limited. Running large scale algorithms becomes challenging because of

these constraints. Most QML research will depend on simulators and small quantum devices until hardware development advances further.

Algorithms are another challenge. Scientists have proposed multiple practical quantum algorithm solutions, yet most of these approaches fail to outperform classical methods when solving problems in the real world.

Open source development also faces multiple obstacles. Different platforms and frameworks lead to platform fragmentation. The process of tool integration becomes challenging because developers struggle to find suitable ecosystems that meet their needs.

In India, one of the challenges is the availability of trained talent. The number of experts in IITs and IISc continues to grow but remains insufficient to meet global market needs. The search for engineers who possess both machine learning expertise and quantum computing abilities proves challenging for most startups. Even in countries with advanced programs, there aren't enough professionals with skills in both domains.

However, quantum hardware companies are actively working to develop stable qubits. Universities have established new interdisciplinary programs that combine physics with computer science and artificial intelligence. Open source communities are uniting various platforms to create enhanced learning experiences. The path to progress is difficult but is showing signs of significant development.

Key open source platforms for QML

Platform	Function	Applications
Qiskit (IBM)	Quantum circuit development and simulation	Widely used for research, education and prototyping
Cirq (Google)	Python-based quantum algorithm development	Suitable for algorithm testing and experimental QML
PennyLane (Xanadu)	Integrates quantum computing with machine learning	Works with PyTorch, TensorFlow; popular for QML research